

CHƯƠNG 9

Trình bày Kết quả Phân tích Dữ liệu

Trí tuệ Nhân tạo cho Kế toán (AI in Accounting)

Khoa Kế toán - Kiểm toán

Bài giảng Thực hành 09

Intro 9.0

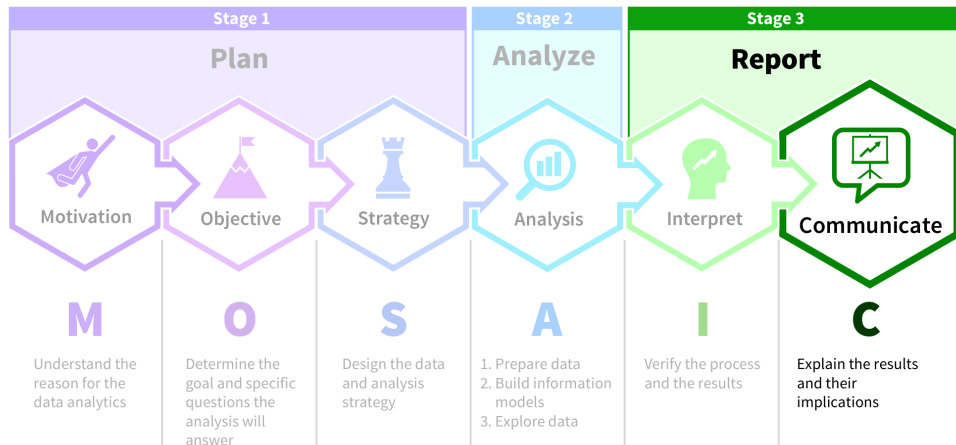


ILLUSTRATION 9.1

Europe	North America	South America
France	Canada	Argentina
Germany	Mexico	Bolivia
Poland	USA	Brazil
Spain		Chile
Sweden		Colombia
United Kingdom		Venezuela

ILLUSTRATION 9.2

Models	Brands		
	Apechete	Jackson	Tatra
	Chare	Brutus	Advantage
	Island	Crux	Bloom
	Pebble	Fiddle	Jespie
	Robin	Rebel	Mortimer
	Summet	Wood	Rambler

ILLUSTRATION 9.3

Gross Sales Trends by Brand: 2024–2025 (in \$ Millions)

	Apechete	Jackson	Tatra
January	\$ 2.4M	\$ 1.2M	\$ 2.5M
February	\$ 4.3M	\$ 2.2M	\$ 2.8M
March	\$ 6.5M	\$ 3.4M	\$ 3.7M
April	\$ 0.2M	\$ 0.2M	\$ 1.3M
May	\$ 2.5M	\$ 1.2M	\$ 2.0M
June	\$ 4.6M	\$ 2.3M	\$ 2.9M
July	\$ 2.3M	\$ 1.2M	\$ 3.2M
August	\$ 2.4M	\$ 1.2M	\$ 3.7M
September	\$ 2.4M	\$ 1.2M	\$ 3.5M
October	\$ 2.6M	\$ 1.7M	\$ 3.2M
November	\$ 2.7M	\$ 1.6M	\$ 2.9M
December	\$ 2.2M	\$ 1.4M	\$ 1.9M

Chương 9: Trình bày Kết quả Phân tích Dữ liệu

- **Mục tiêu:** Xây dựng năng lực phân tích dữ liệu và trình bày kết quả cho kế toán viên.
- **Tầm quan trọng:** Đóng vai trò là cầu nối giữa dữ liệu kỹ thuật và quyết định chiến lược.
- **Lộ trình:** Từ định nghĩa câu chuyện dữ liệu đến tạo trực quan hóa và nhận diện các biểu đồ gây hiểu lầm.

ILLUSTRATION 9.4

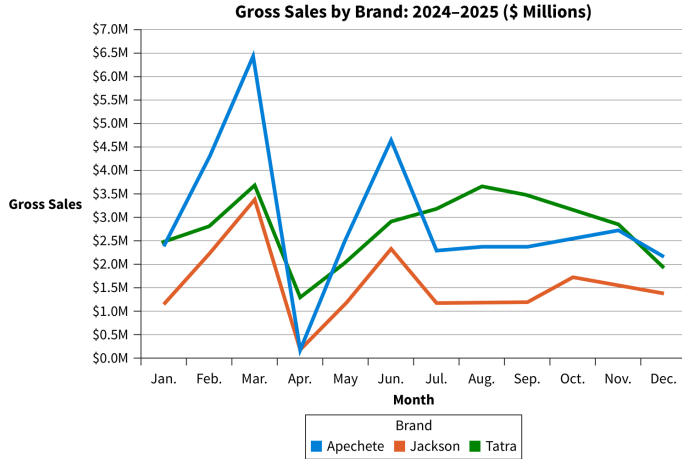


ILLUSTRATION 9.5

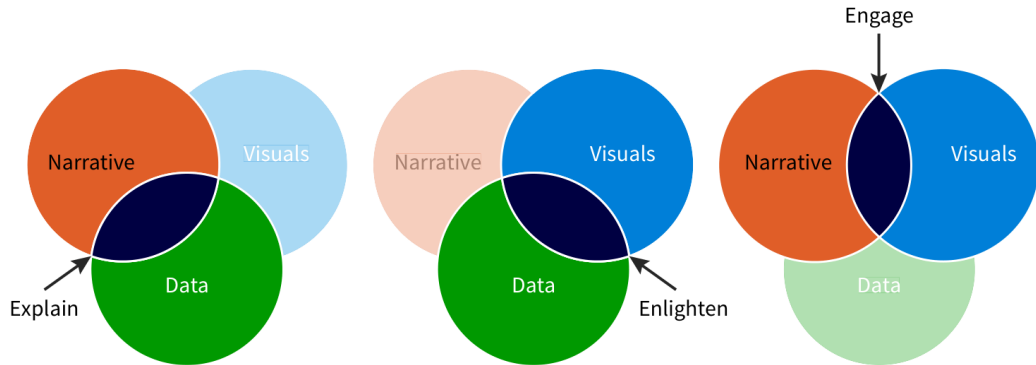
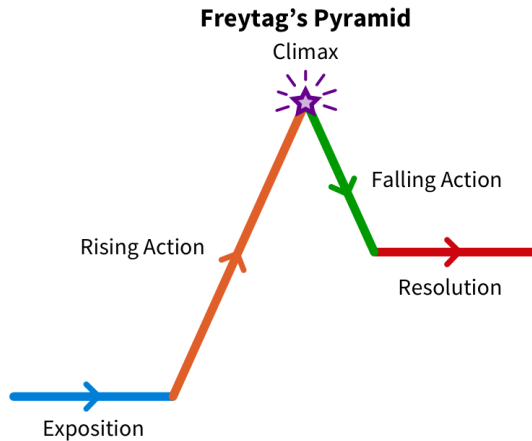


ILLUSTRATION 9.6



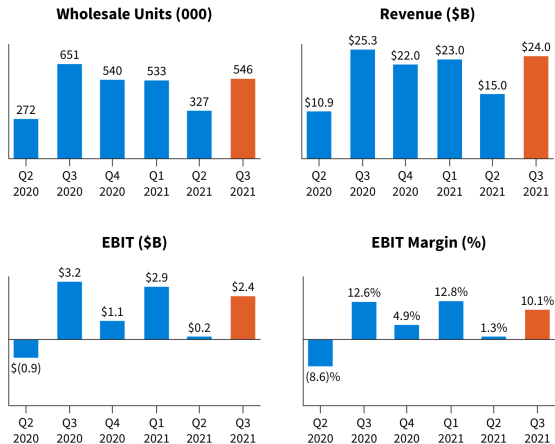
9.1 Kể Chuyện Bằng Dữ Liệu

- **Câu chuyện dữ liệu (Data Story) là gì?** Là sự giao thoa giữa dữ liệu, hình ảnh trực quan, và tường thuật logic.
- **Mục đích:** Giúp chuyển hóa những bảng số liệu khô khan thành các insight có thể hành động.
- Không chỉ là vẽ biểu đồ, mà là việc trả lời câu hỏi: *"Dữ liệu này có ý nghĩa gì đối với công ty?"*

ILLUSTRATION 9.7

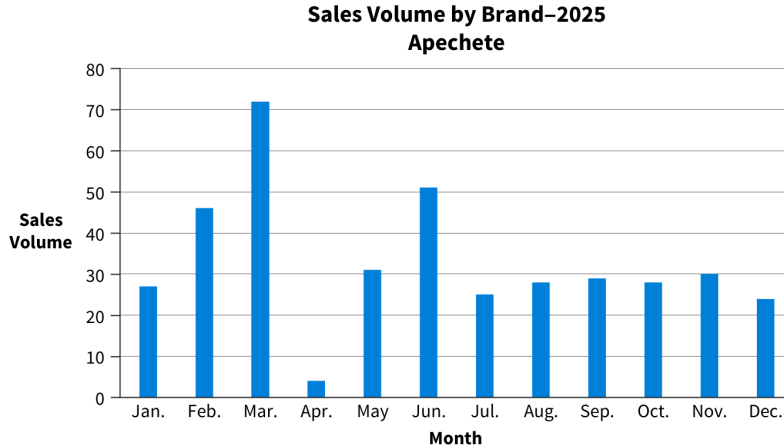
Freytag's Pyramid	Data Story Application	Kickback Analysis Example
Exposition	Introduce the problem or issue.	Briefly discuss the background information relevant to the analysis. Include interesting details to engage the reader's attention. Example: "Is kickback fraud occurring in the Purchasing Department? Based on an anonymous tip, we took a closer look at the purchasing department."
Rising Action	The subject of the analysis is explored at a deeper level.	In this part of the story, methodically peel back layers of the kickback analyses.
Climax	The main finding or insight is shared. This is the "aha moment" of the story.	After building the case in the previous section, announce the suspected employee(s) and vendor(s).
Falling Action	Share the solution.	Identifying a suspected employee and vendor does not prove kickback fraud has occurred. Next, provide more details and recommendations as to specific transactions that should be investigated further.
Resolution	Conclude the story and provide next steps.	Make suggestions for additional internal controls to avoid future kickback fraud.

ILLUSTRATION 9.8



Source: Ford, Q3 2021 Earnings Review, October 27, 2021.

ILLUSTRATION 9.9



Phát triển Năng lực Dữ liệu

- **Data Literacy:** Khả năng đọc, làm việc, phân tích và tranh luận bằng dữ liệu.
- **Vai trò của Kế toán:** Là chuyên gia am hiểu hoạt động tài chính, kế toán viên ở vị thế lý tưởng nhất để trở thành "người kể chuyện" bằng dữ liệu cho Ban Giám đốc.

ILLUSTRATION 9.10

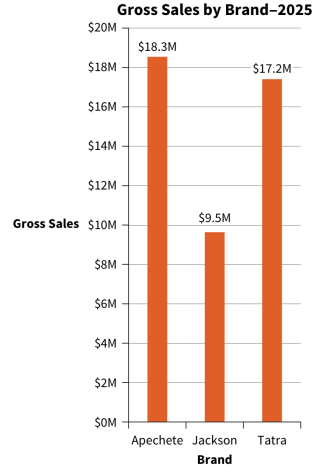
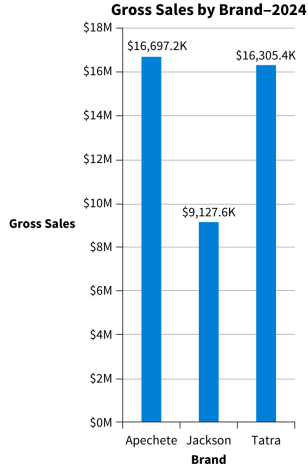
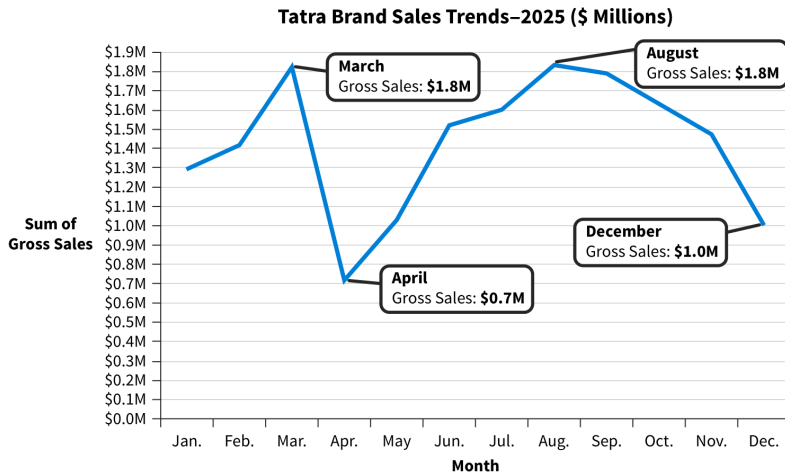


ILLUSTRATION 9.11

Category	Description	What They Want
Novice	Has never encountered the information.	Enough detail to gain understanding.
Managerial	Has some knowledge of the topic.	Actionable results.
Expert	Has deep knowledge of the topic.	Investigation and discovery.
Executive	Has a broad, high-level knowledge of the topic.	Only the most important insights.

ILLUSTRATION 9.12



Kỹ thuật Xây dựng Câu chuyện

- **Bối cảnh (Context):** Dữ liệu này được thu thập khi nào? Nó phản ánh xu hướng gì?
- **Giao tiếp hiệu quả (Communicate Effectively):** Hãy trình bày rõ ràng thông điệp, đừng bắt khán giả phải tự tìm câu trả lời trong dữ liệu.
- **Xác định cấu trúc cốt truyện:** Bắt đầu bằng thực trạng, chỉ ra nút thắt (vấn đề) và đề xuất giải pháp.

ILLUSTRATION 9.13

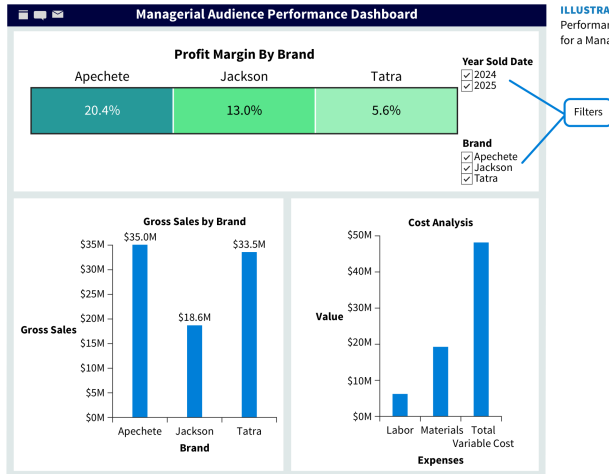


ILLUSTRATION 9.13

Performance Dashboard Sample
for a Managerial Audience

ILLUSTRATION 9.14

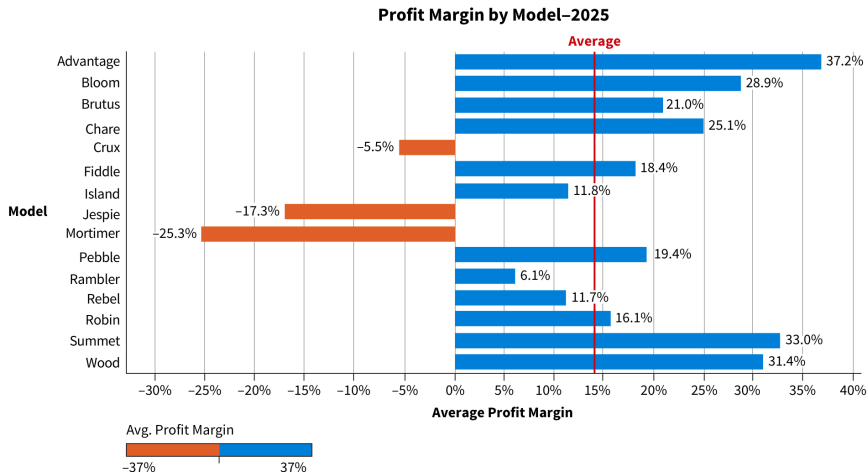
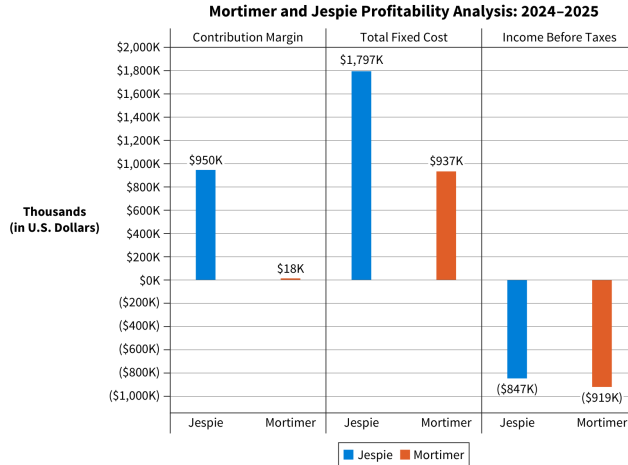


ILLUSTRATION 9.15



9.2 Quy Trình Tạo Trực Quan Hóa

- Trực quan hóa dữ liệu hiệu quả cần một quy trình tuần tự.
- **Quy trình 3 bước cốt lõi:**
 - 1 Xác minh Dữ liệu (Verify the Data)
 - 2 Thấu hiểu Khán giả (Consider the Audience)
 - 3 Xác định Mục tiêu (Define the Objective)

ILLUSTRATION 9.16

Objective	Explanation	Example
Composition	Show how part of the data compares to the whole.	How much revenue has each region contributed to total revenue?
Relationships	Show how the data are related.	Is there a relationship between machine hours and maintenance expense?
Distributions	Reveal how data are spread out or grouped.	Are there transactions that might be considered outliers?
Trends	Display patterns in the data.	Is there a seasonal pattern in revenue?
Comparisons	Compare values between groups of data.	How does the revenue for 2024 compare to 2025 by product?

ILLUSTRATION 9.17

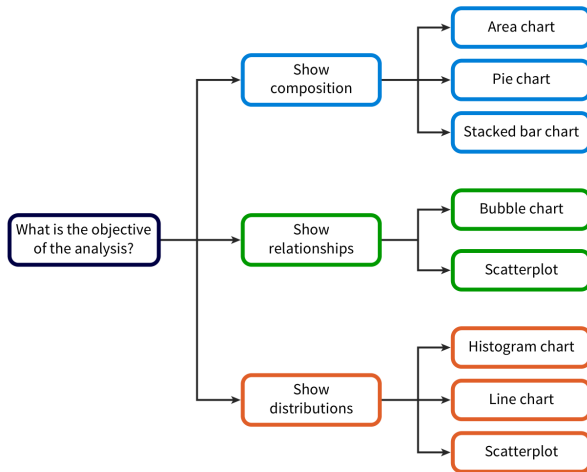
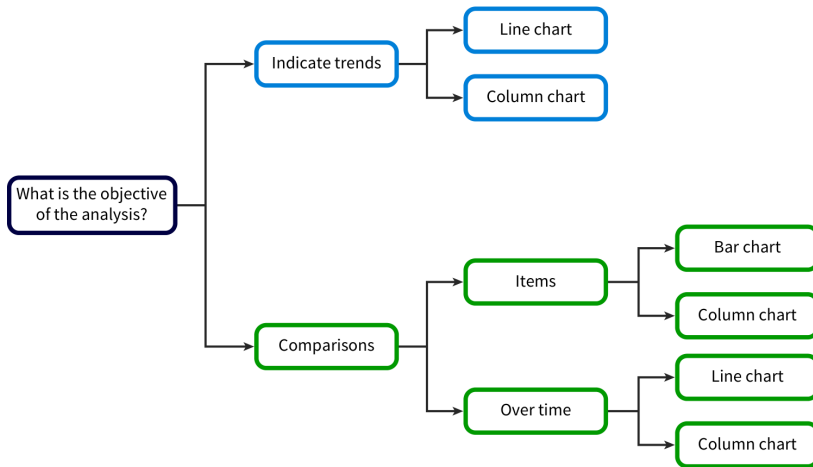


ILLUSTRATION 9.18



Bước 1 & 2: Xác minh & Thẩm hiểu Khán giả

- **Xác minh Dữ liệu:** Đảm bảo tính chính xác, đầy đủ và hợp lệ. Dữ liệu sai sẽ dẫn đến biểu đồ sai (Garbage In, Garbage Out).
- **Thẩm hiểu Khán giả:** Ai sẽ xem biểu đồ này? Họ có kiến thức chuyên môn về tài chính hay không? Mức độ chi tiết họ cần là bao nhiêu?

ILLUSTRATION 9.19

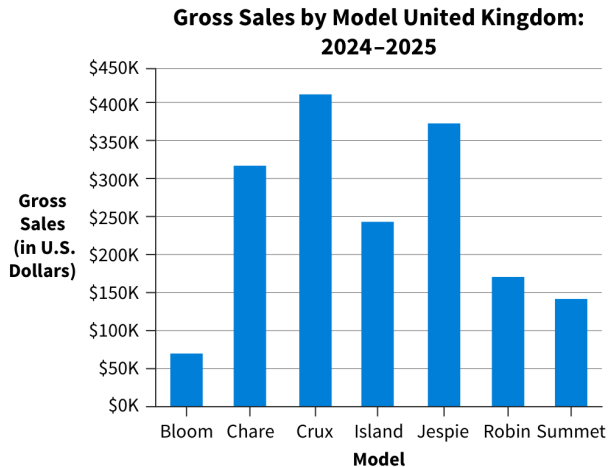


ILLUSTRATION 9.20

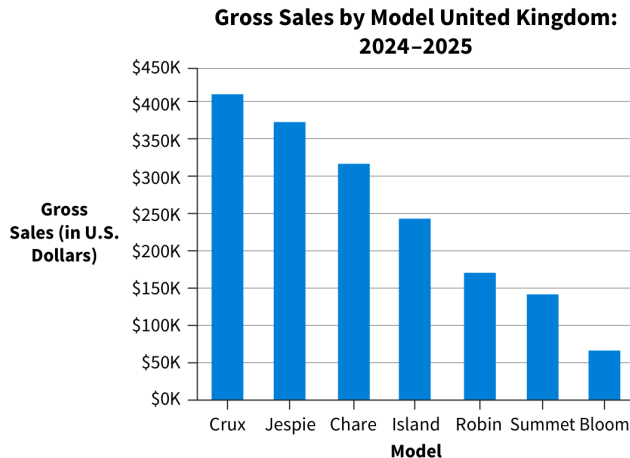
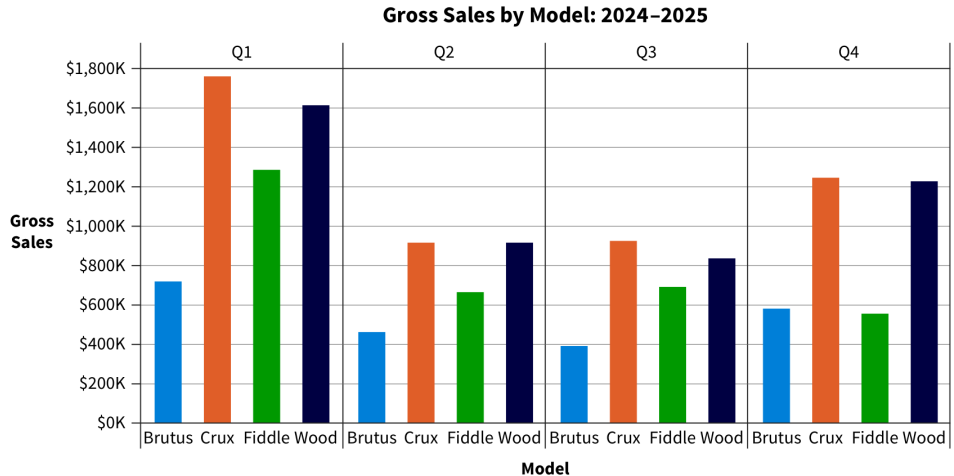


ILLUSTRATION 9.21



Bước 3: Xác định Mục tiêu (Define the Objective)

- **Mục tiêu:** Biểu đồ nhằm mục đích so sánh, chỉ ra xu hướng, hay hiển thị tỷ trọng?
- **Lựa chọn biểu đồ (Chart selection):** Phải chọn đúng biểu đồ phù hợp với mục tiêu.
 - *Ví dụ:* So sánh giữa các mục thì dùng Bar Chart. Xem xu hướng theo thời gian thì dùng Line Chart.

ILLUSTRATION 9.22

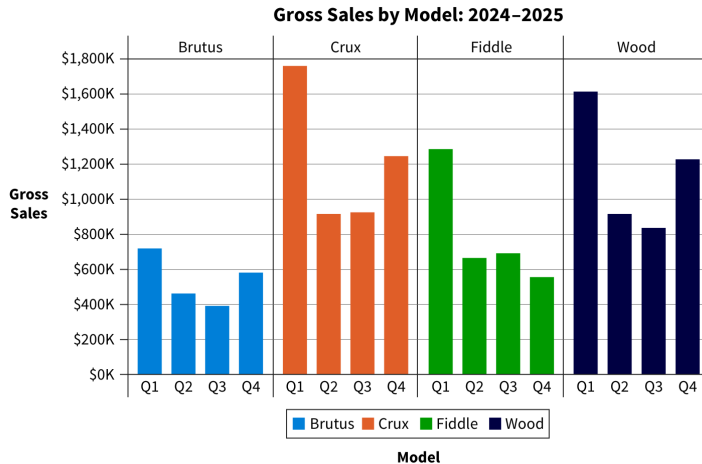


ILLUSTRATION 9.23

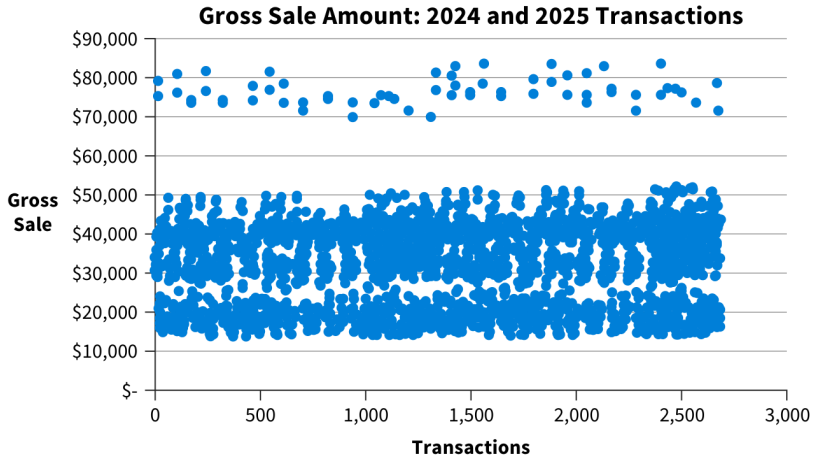
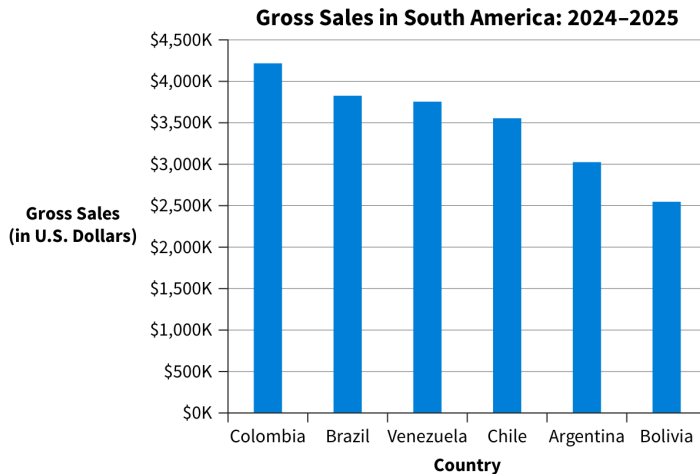


ILLUSTRATION 9.24



9.3 Đặc Điểm Trực Quan Hóa Hiệu Quả

- **Hiệu quả là gì?** Là khả năng truyền tải thông điệp nhanh nhất và ít gây nhầm lẫn nhất.
- **Nhận thức Thị giác (Visual Perception):** Não bộ con người được tối ưu hóa để xử lý hình ảnh và màu sắc thay vì các con số dạng văn bản.
- **Chìa khóa:** Khai thác sức mạnh của hệ thống thị giác để hướng sự chú ý của người xem.

ILLUSTRATION 9.25

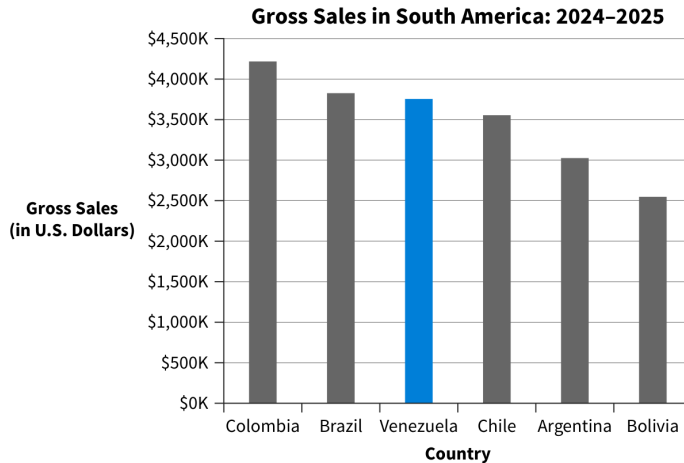


ILLUSTRATION 9.26

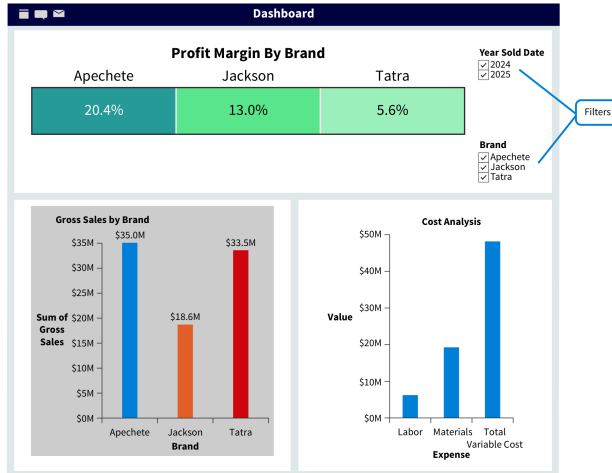


ILLUSTRATION 9.27

SUM (Gross Sales)



Các Thuộc tính Tiền-chú-ý (Preattentive Attributes)

- **Định nghĩa:** Các đặc điểm mà não bộ xử lý vô thức trước cả khi ta tập trung suy nghĩ.
- **Màu sắc (Color):** Sử dụng màu đỏ để nhấn mạnh sự bất thường hoặc thua lỗ.
- **Kích thước & Hình dạng (Size & Shape):** Dùng điểm lớn hơn hoặc đường nét đậm hơn để nhấn mạnh dữ liệu quan trọng nhất.

ILLUSTRATION 9.28

Profit Ratio



ILLUSTRATION 9.29

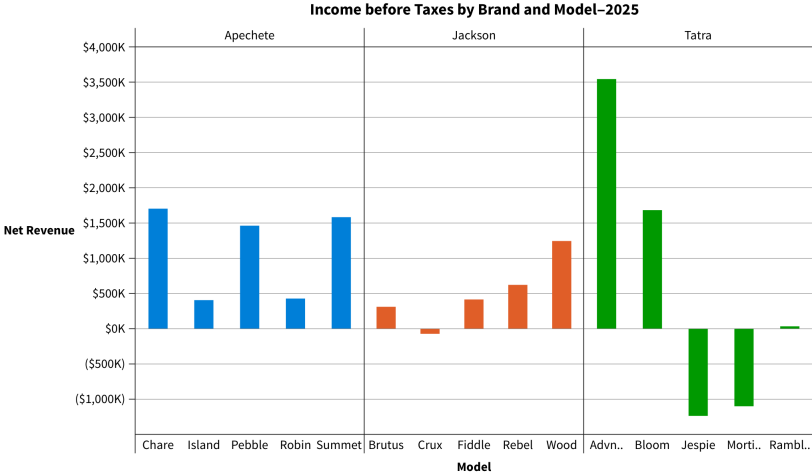
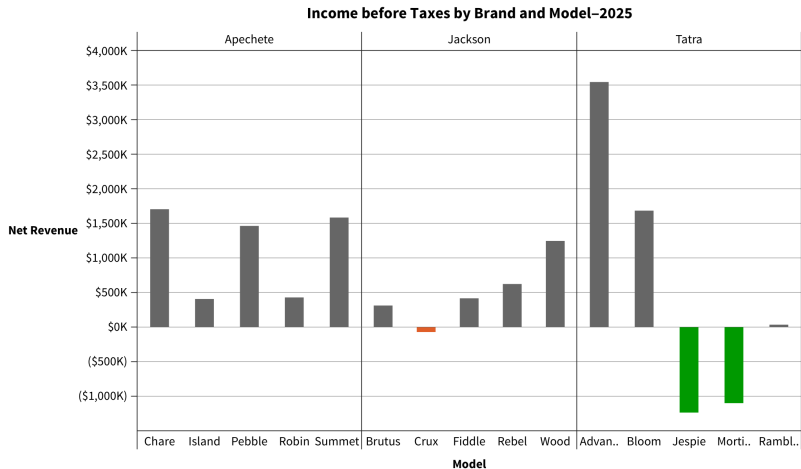


ILLUSTRATION 9.30



Tối Giản Hóa và Tránh Lộn Xộn

- **Avoid Clutter:** Tránh sự rối rắm không cần thiết.
- **Chartjunk:** Các thành phần trang trí không mang lại giá trị thông tin (VD: Hiệu ứng 3D, lưới nền quá đậm, màu sắc sặc sỡ không cần thiết).
- Tối giản hóa giúp thông điệp chính nổi bật (Nguyên lý Tỷ lệ Dữ liệu/Mực in - Data-ink ratio).

ILLUSTRATION 9.31

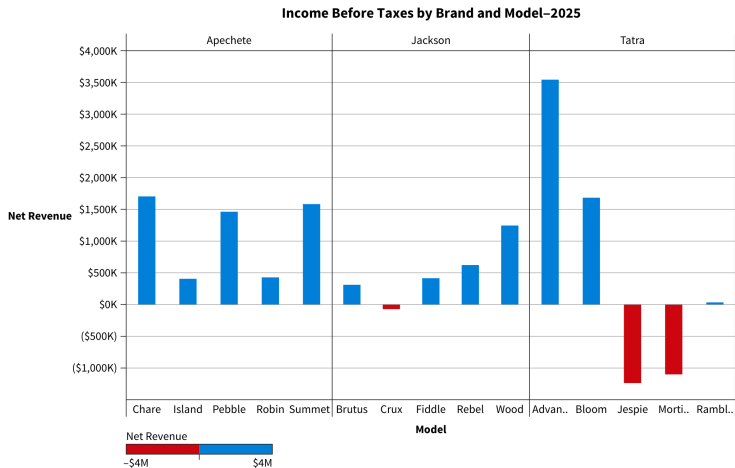


ILLUSTRATION 9.32

Income before Taxes by Region			
Region	Apechete	Jackson	Tatra
Europe	\$ 1,246,353	\$ 1,010,446	\$ 136,006
North America	\$ 2,013,345	\$ 963,145	\$ 2,719,779
South America	\$ 2,372,282	\$ 625,278	\$ 124,690

Net Revenue



\$124,690

\$3M

ILLUSTRATION 9.33

Income before Taxes by Region			
Region	Apechete	Jackson	Tatra
Europe	\$ 1,246,353	\$ 1,010,446	\$ 136,006
North America	\$ 2,013,345	\$ 963,145	\$ 2,719,779
South America	\$ 2,372,282	\$ 625,278	\$ 124,690

Income before Taxes



9.4 Nhận Diện Trực Quan Hóa Gây Hiểu Lầm

- Dữ liệu có thể không biết nói dối, nhưng biểu đồ thì có thể (Misleading Data Visualizations).
- Việc thiết kế sai nguyên tắc có thể bóp méo góc nhìn của người xem.
- Kế toán viên cần tinh ý nhận diện những cạm bẫy thiết kế phổ biến.

ILLUSTRATION 9.34

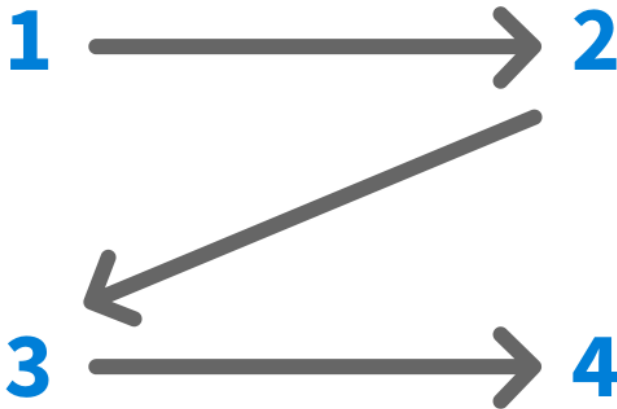


ILLUSTRATION 9.35

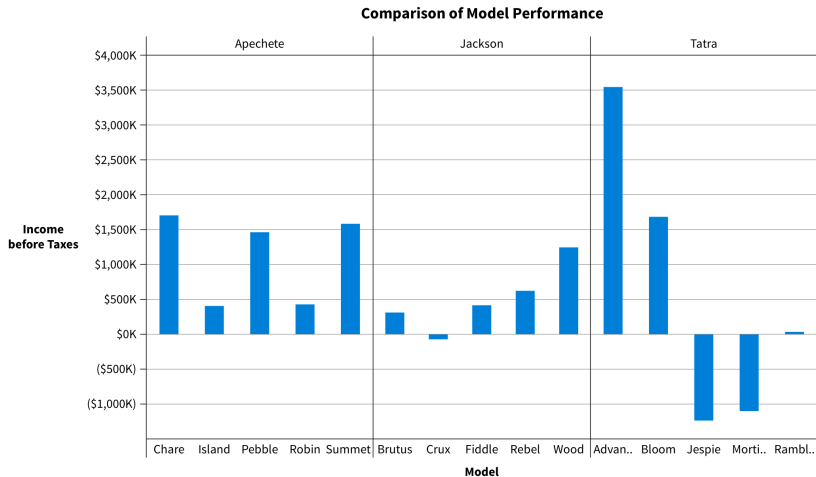
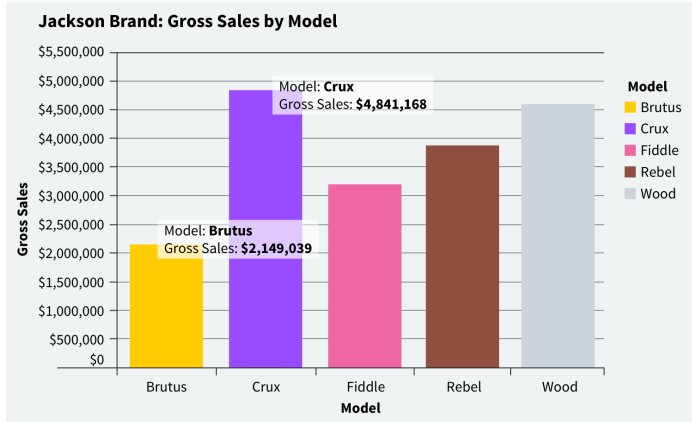


ILLUSTRATION 9.36

HMC Gross Sales Jackson Models: 2024–2025



Lỗi 1: Bỏ qua đường cơ sở (Omitting Baseline)

- Đối với biểu đồ cột (Bar chart), trục Y **bắt buộc** phải bắt đầu từ số 0.
- **Hậu quả:** Nếu trục Y bị cắt xén, một sự chênh lệch nhỏ sẽ bị khuếch đại thành một sự thay đổi khổng lồ.

ILLUSTRATION 9.37

Jackson Brand: Gross Sales by Model: 2024–2025

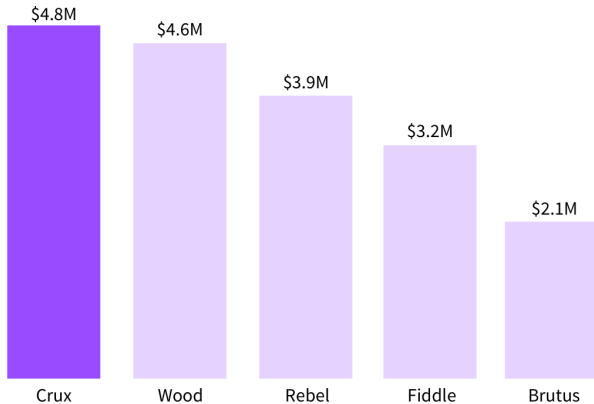


ILLUSTRATION 9.38

Cluttered Visualization	Revised Visualization
Too many colors are not adding to the meaning of the graph because the models are also directly labeled.	The visual is a single color.
Axis labels are provided, but the audience must guess the exact amount.	Label each bar directly with the sales volume amount and remove the axis. Sales volume is in the title of the graph, so remove the axis header.
Annotation text boxes identify the bestselling model.	Sort the data from highest to lowest.
Background color of the graph is unnecessary.	Remove background color. Remove gridlines since we are removing the axis and labeling the columns directly.
Models are listed in alphabetical order. There is no indication which model the audience should focus on.	The bestselling model is highlighted and the others are grayed out to draw attention to it.

ILLUSTRATION 9.39

1. Data have been verified.	11. Graph does not have border line (reducing clutter).
2. Visualization addresses the objective or question being addressed in the analysis.	12. Data are labeled directly rather than in a separate legend.
3. It is appropriate for the intended audience.	13. Redundant labels are removed.
4. The type of visual is appropriate for the data and the level of precision needed.	14. Y-axis scales start at zero.
5. A descriptive title is left-justified in the upper left corner.	15. Proportions are accurate.
6. Subtitle and/or annotations provide additional information.	16. Data are intentionally ordered.
7. Significant findings or conclusions are highlighted.	17. Display is free from distractions.
8. Text size is hierarchical and readable.	18. Color is used sparingly and consistently.
9. Text is horizontal when possible.	19. Color is legible for people with colorblindness.
10. Gridlines, if used, are muted.	20. Spatial flow is intuitive for the viewer.

Lỗi 2: Thao Túng Trục Y

- **Đảo ngược trục (Inverted Y-axis):** Gây cảm giác lầm tưởng rằng số liệu đang tăng trong khi thực tế đang giảm.
- **Chia tỷ lệ không đều (Uneven scaling):** Trục tọa độ giãn cách không đều làm đường xu hướng bị biến dạng.

ILLUSTRATION 9.40A

Visualization	Best Practices
Area Chart 	- Do not use with more than four categories to avoid confusion and clutter. - Start the y-axis at zero. - Put highly variable data on the top and data with low variability on the bottom.
Bar Chart 	- Use horizontal bars if there are more than 7 categories or long category labels. - Use horizontal labels for better readability. - Space bars appropriately and consistently. - Use color sparingly, or as an accent. - Always have a zero baseline (the y-axis begins at zero). - Compare 2-7 categories with vertical columns.
Column Chart 	
Bubble Chart 	- Label bubbles and make sure they are visible. - Scale bubble size by area and not diameter. - Do not use bubbles if they are all similar in size.
Histogram Chart 	- Use a zero baseline. - Choose an appropriate number of bins: - Bins are numbers that represent the intervals into which the data will be grouped. - Bins define the groups used for the frequency distribution. - Generally, include between 5-15 bins.
Line Graph 	- Time runs from left to right. - Be consistent when plotting time points. - Use solid lines, not dotted. - Use a zero baseline. - Do not plot more than four lines. Use multiple charts, instead.
Pie Chart 	- Most impactful with small data sets. - Best to use when showing differences within groups based on one variable. - Ensure the data adds to 100%. - Limit the chart to a maximum of five segments. - Start the first segment at 12 o'clock position.
Stacked Bar Chart 	- Can be vertical or horizontal. - Follow same best practices as bar charts. - Used to show comparisons of subcomponents between categories. - Best used when there are not too many subcomponents. - Consider using a 100% stacked bar to make comparisons between the bars and subcomponents easier.

ILLUSTRATION 9.40B

Visualization

Best Practices

Scatter Chart



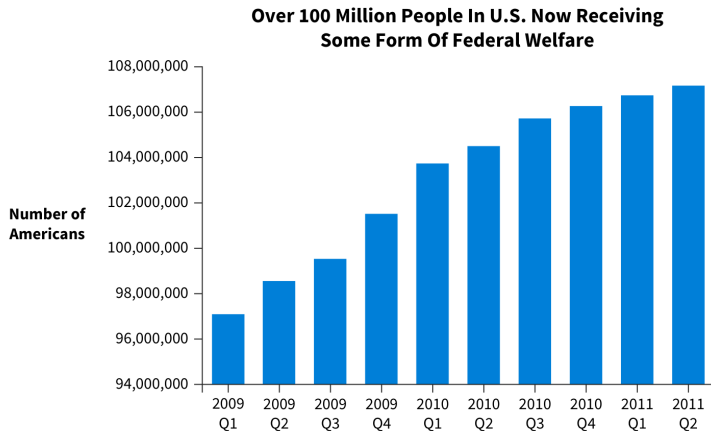
- Data set should be in pairs with an independent variable (x-axis) and a dependent variable (y-axis).
- Use if order is not relevant—otherwise use a line graph.
- Do not use if there are only a few pieces of data or if there is no correlation.

Tree Map



- Appropriate when precise comparisons are not important.
- Use bright, contrasting colors so that each box is easily defined.
- Label boxes with text or numbers.

ILLUSTRATION 9.41



Source: US census survey / U.S Department of Commerce / Public Domain.

Lỗi 3: Đi Ngược Quy Ước Thiết Kế

- **Going Against Conventions:** Con người có các quy ước ngầm về màu sắc và không gian.
- *Ví dụ:* Màu Đỏ thường mang ý nghĩa tiêu cực/lỗi, màu Xanh mang ý nghĩa tích cực/lãi. Dùng ngược lại sẽ gây nhầm lẫn trầm trọng.
- Trên/Phải thường mang ý nghĩa cao hơn/tăng lên. Dưới/Trái là giảm.

ILLUSTRATION 9.42

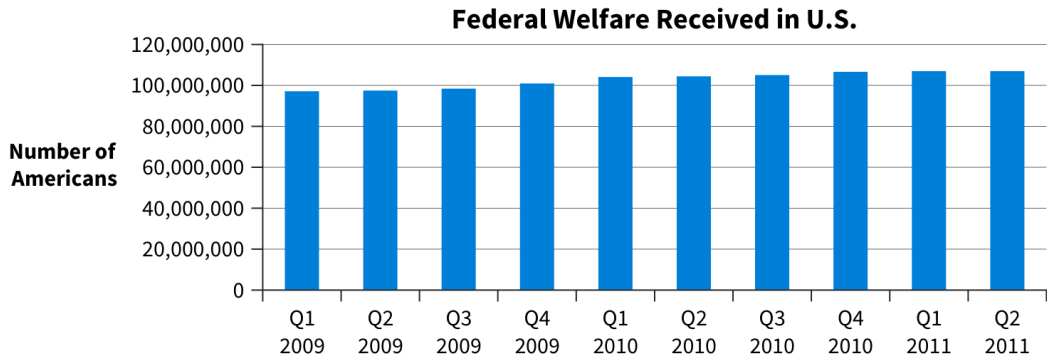


ILLUSTRATION 9.43

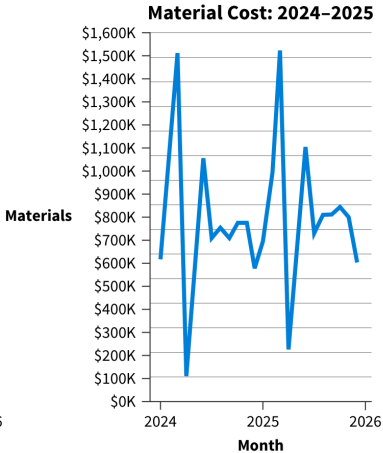
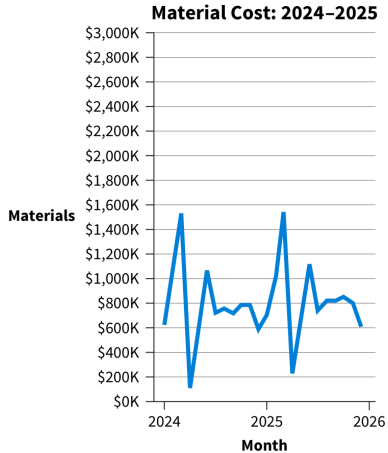
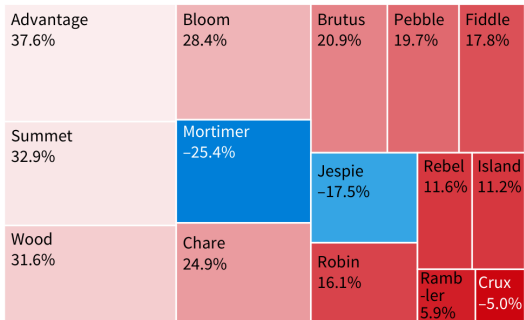
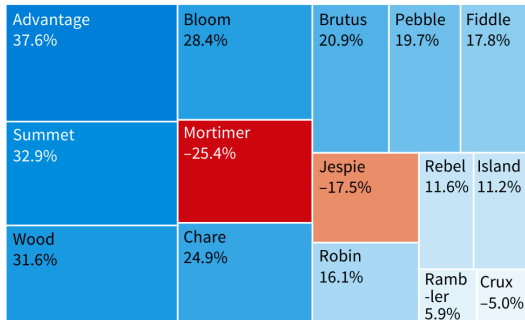


ILLUSTRATION 9.44

Profit Margin by Model (Misleading)



Profit Margin by Model (Accurate)



Lỗi 4: Chọn Lọc Dữ Liệu & Dùng Sai Biểu Đồ

- **Cherry-picking:** Chỉ chọn hiển thị khoảng thời gian dữ liệu có lợi để che giấu bức tranh tổng thể.
- **Wrong Graph:** Dùng sai biểu đồ, ví dụ dùng Pie Chart (Biểu đồ tròn) cho dữ liệu tổng không bằng 100%, hoặc dùng quá nhiều lát cắt.

ILLUSTRATION 9.45

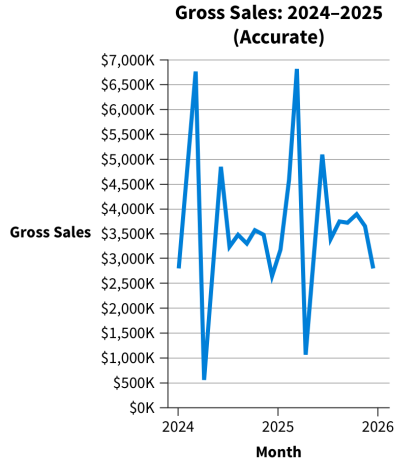
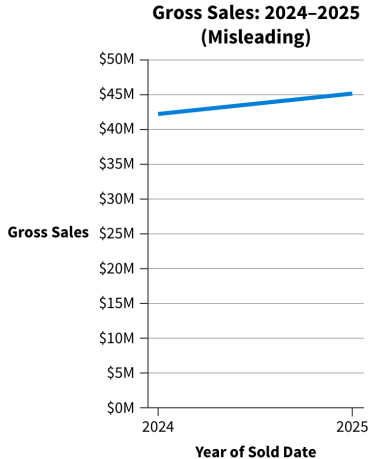
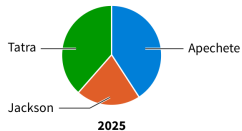
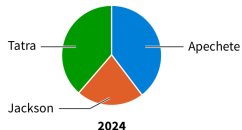


ILLUSTRATION 9.46

**Brand Sales Comparison
(Misleading)**



Brand Sales Comparison (Accurate)

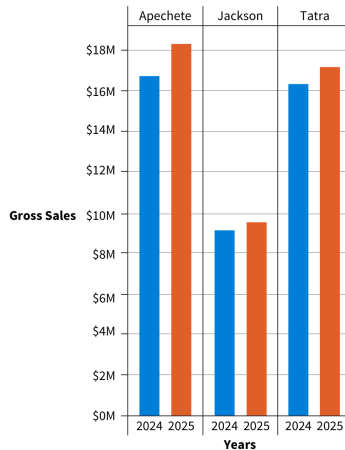


ILLUSTRATION 9.47

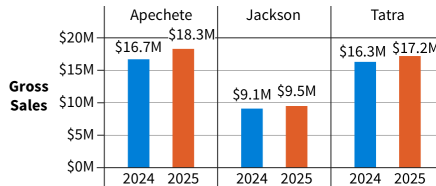
Gross Sales by Region

Region	2024	2025
Europe	\$5.0M	\$12.6M
South America	\$9.7M	\$12.8M
North America	\$27.4M	\$19.5M

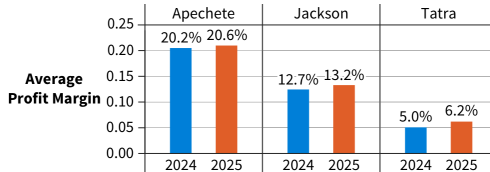
Gross Sales



Gross Sales by Brand



Profit Margin by Brand



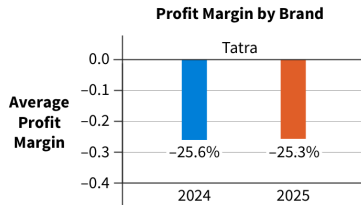
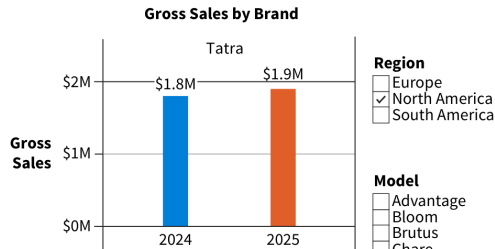
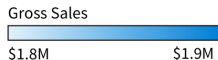
9.5 Thiết Kế Bài Thuyết Trình Tương Tác

- Việc báo cáo bằng file tĩnh truyền thống (PDF, Excel) đang dần nhường chỗ cho các báo cáo tương tác.
- **Interactive Dashboards:** Người dùng có thể click, filter (lọc) và drill-down (đi sâu) để xem dữ liệu theo ý muốn.

ILLUSTRATION 9.48

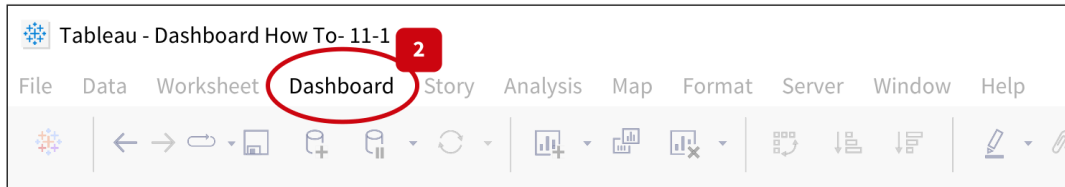
Gross Sales by Region

Region	2024	2025
North America	\$1.8M	\$1.9M



- Model**
- ☐ Advantage
 - ☐ Bloom
 - ☐ Brutus
 - ☐ Chare
 - ☐ Crux
 - ☐ Fiddle
 - ☐ Island
 - ☐ Jespie
 - ☒ Mortimer
 - ☐ Pebble
 - ☐ Rambler
 - ☐ Rebel
 - ☐ Robin
 - ☐ Summit
 - ☐ Wood

ILLUSTRATION 9.49



- **Best Practices for Live Presentations:**

- Không đưa quá nhiều chữ lên slide (khán giả sẽ đọc chữ thay vì nghe bạn nói).
- Sử dụng biểu đồ để hỗ trợ lập luận. Giải thích rõ các trục tung/hoành trước khi phân tích xu hướng.

ILLUSTRATION 9.50

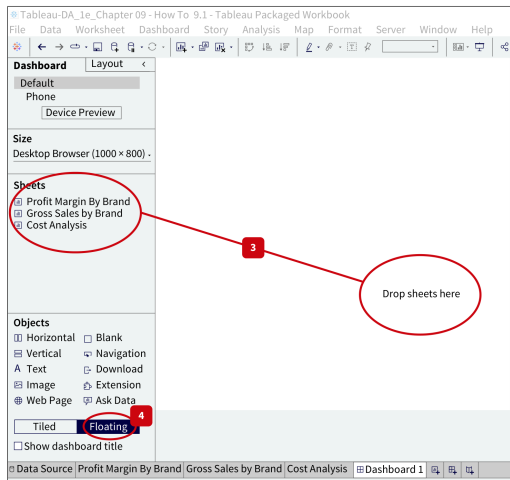
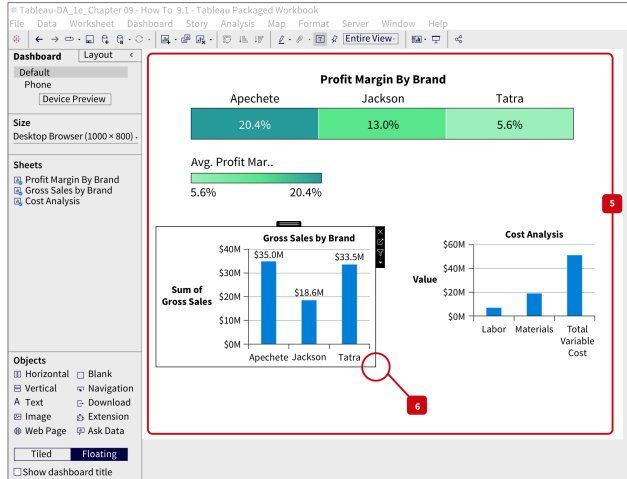


ILLUSTRATION 9.51



Case Study: Báo Cáo Kế Toán Trực Quan

- Ứng dụng xây dựng Dashboard trong các cơ quan nhà nước, thư viện công cộng (VD: Madison Public Library) và doanh nghiệp.
- Cho phép Ban Giám đốc nhìn được bức tranh thu-chi và phân bổ ngân sách real-time.

ILLUSTRATION 9.52

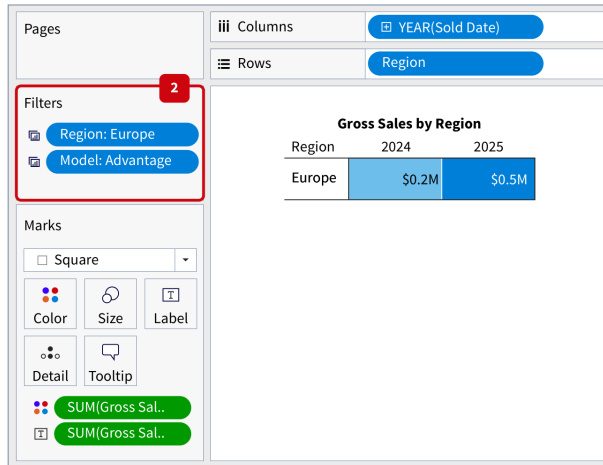
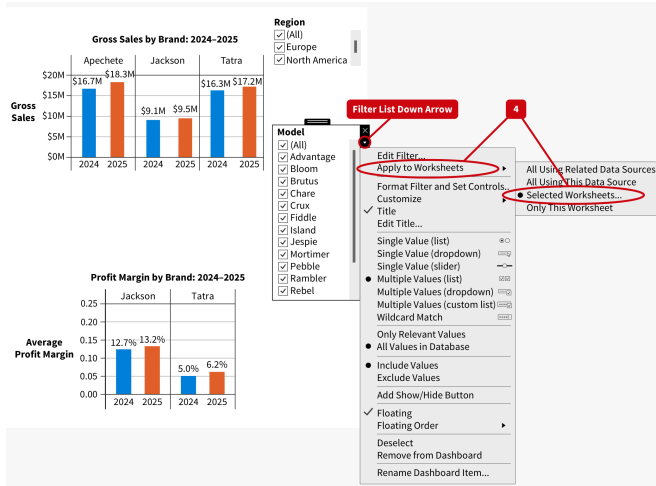


ILLUSTRATION 9.53A



- **Tóm tắt LO 9.1 - LO 9.5:** Kỹ năng phân tích dữ liệu không chỉ nằm ở việc xử lý kỹ thuật, mà quan trọng nhất là **truyền đạt (communicate)** được giá trị của dữ liệu.
- Trực quan hóa đúng cách giúp tối ưu hóa ra quyết định.

ILLUSTRATION 9.53B

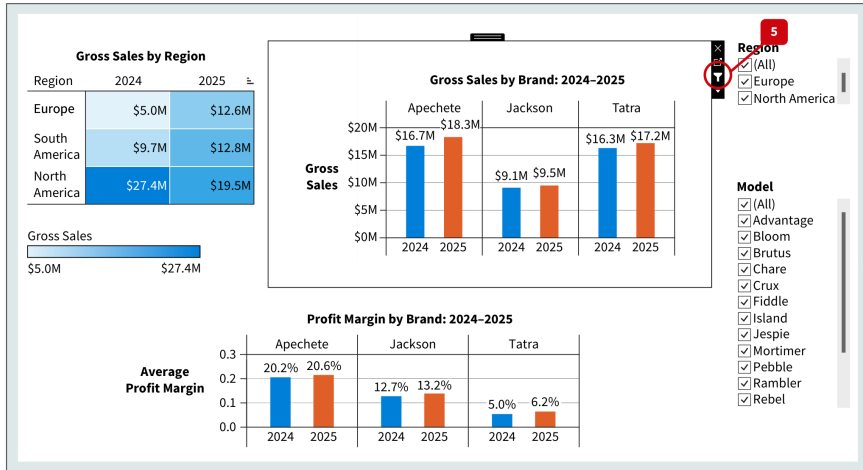
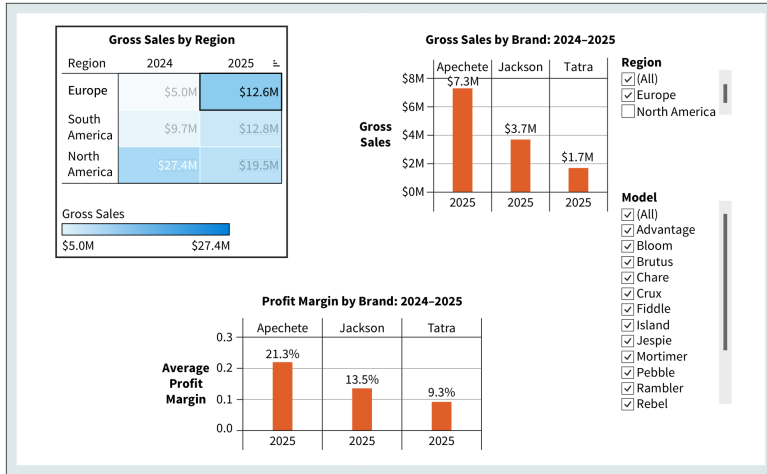
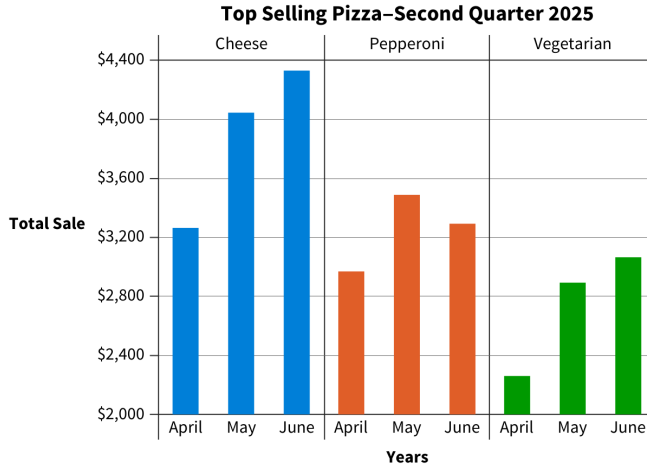
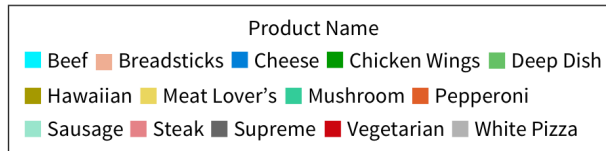
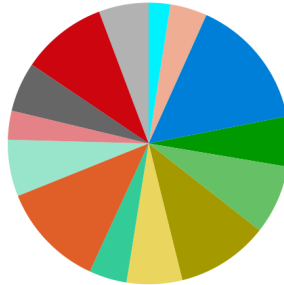


ILLUSTRATION 9.55







Profit Margin By Brand–2025

Apechete

Jackson

Tatra

20.4%

13.0%

5.6%

Apply It 9.1

**U.S. Outdoor Adventures
2025 Sales by Segment**

Segment	Total Sales
Consumer	\$ 512,711
Corporate	\$ 359,264
Travel Agency	\$ 232,095
Grand Total	\$ 1,104,070

U.S. Outdoor Adventures Products Offered

Camping Gear	Paddle	Tents
Camp stove	Kayak	Backpacking Tent
Chairs	Life Vest	Base Camp Model
Cook set	Paddle Board	Mountain Tent - 4 Person
Fasteners	Paddles	Northface subzero
Firestarter Kit		
First Aid Kit		
Micro Cooking unit		
Propane Portable Grill		
Sleeping Bag		

U.S. Outdoor Adventures 2025 Financial Information

	Camping Gear	Paddle	Tents	Total
Sales	\$ 321,964	\$ 243,580	\$ 538,526	\$ 1,104,070
Discount	\$ 34,170	\$ 40,030	\$ 96,864	\$ 171,063
Net Sales	\$ 287,794	\$ 203,550	\$ 441,663	\$ 933,006
Cost of Goods Sold	\$ 114,903	\$ 87,682	\$ 193,681	\$ 396,265
Shipping Cost	\$ 36,900	\$ 31,066	\$ 64,646	\$ 132,611
Profit	\$ 135,992	\$ 84,802	\$ 183,336	\$ 404,130

Madison public library 9.1

Library Name	Street Address	City	State	Zip Code
Alicia Ashman	733 N High Point Rd	Madison	WI	53717
Central	201 W Mifflin St	Madison	WI	53703
Hawthorne	2707 E Washington Ave	Madison	WI	53704
Lakeview	2845 N. Sherman Ave	Madison	WI	53704
Meadowridge	5726 Raymond Rd	Madison	WI	53711
Pinney	516 Cottage Grove Rd	Madison	WI	53716
Sequoia	4340 Tokay Blvd	Madison	WI	53711
South Madison	222 S Park St	Madison	WI	53713

Apply It 9.2

Audit Test	Possible Visualization
Comparison of units sold and amounts billed for every sales transaction.	
Sales revenue by quarter and year.	
Number of tents sold by tent type for current year and three prior years.	
Discount offered per transaction to identify unusual discounts.	
Actual versus expected revenue.	
Changes in general ledger account balances.	

SOLUTION

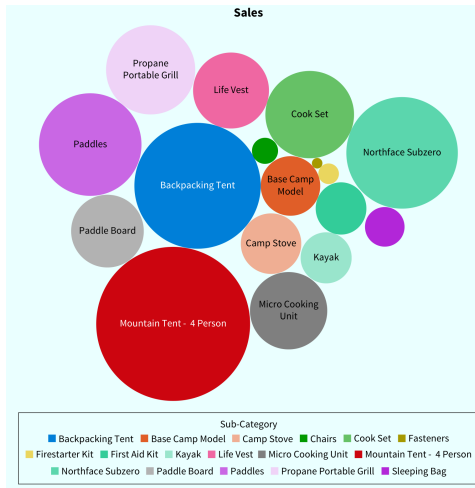
Audit Test	Possible Visualization
Comparison of units sold and amounts billed for every sales transaction.	Scatterplot
Sales revenue by quarter and year.	Column chart or stacked column chart
Number of tents sold by tent type for current year and three prior years.	Stacked column chart or separate line charts for each product
Discount offered per transaction to identify unusual discounts.	Scatterplot
Actual versus expected revenue.	Line chart
Changes in general ledger account balances.	Bar chart

Madison public library 9.2

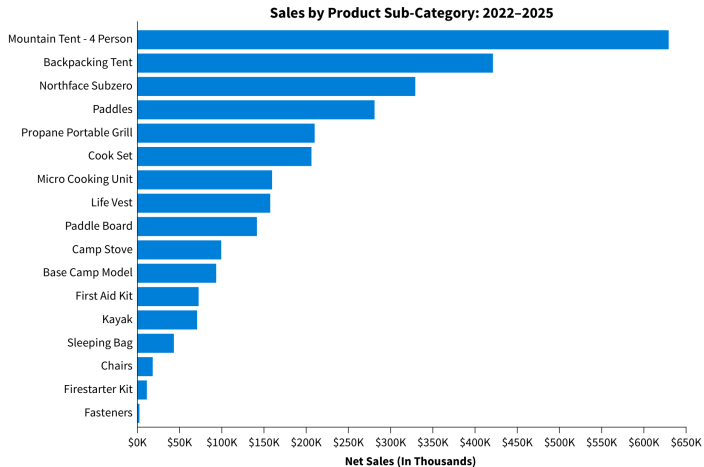
Library Financial Information

	Year Ended December 31			
Library Revenues	2025	2024	2023	2022
City of Madison Library Appropriation	\$ 17,703,566	\$ 17,779,030	\$ 16,915,564	\$ 16,288,835
South Central Library System Contractual Services	\$ 395,478	\$ 337,246	\$ 337,361	\$ 356,336
Dane County Library System Contractual Services	\$ 1,144,935	-	-	-
LINK Contractual Services	\$ 404,255	\$ 454,290	\$ 454,255	\$ 454,255
Fines and Fees	\$ 335,984	\$ 383,403	\$ 395,421	\$ 404,399
Grants	\$ 602,994	\$ 149,459	\$ 1,010,390	\$ 370,254
Other	\$ 74,538	\$ 121,886	\$ 104,631	\$ 124,395
Endowment	\$ 20,000	\$ 20,000	\$ 20,000	\$ 20,000
	\$ 20,681,750	\$ 19,245,314	\$ 19,237,622	\$ 18,018,474
Library Expenses				
Salaries and Benefits	\$ 13,026,440	\$ 12,659,647	\$ 12,352,852	\$ 11,474,221
Library Books, Media, and Databases:	\$ 1,040,746	\$ 1,039,586	\$ 1,000,816	\$ 1,046,644
Dane County Library System Contractual Services	\$ 1,537,180			
Facilities	\$ 1,227,112	\$ 1,515,114	\$ 1,456,628	\$ 1,357,358
LINKcat Online Computer Operations	\$ 623,845	\$ 609,444	\$ 611,337	\$ 592,158
Debt Retirement	\$ 2,826,376	\$ 2,648,112	\$ 2,745,463	\$ 2,720,545
Supplies and Capital Assets	\$ 482,606	\$ 497,976	\$ 390,440	\$ 330,283
Purchased Services, Miscellaneous	\$ 541,895	\$ 745,755	\$ 578,811	\$ 604,312
	\$ 21,306,200	\$ 19,715,634	\$ 19,136,347	\$ 18,125,521
Net Inflow/(Outflow)	\$ (624,450)	\$ (470,320)	\$ 101,275	\$ (107,047)

Apply It 9.3A



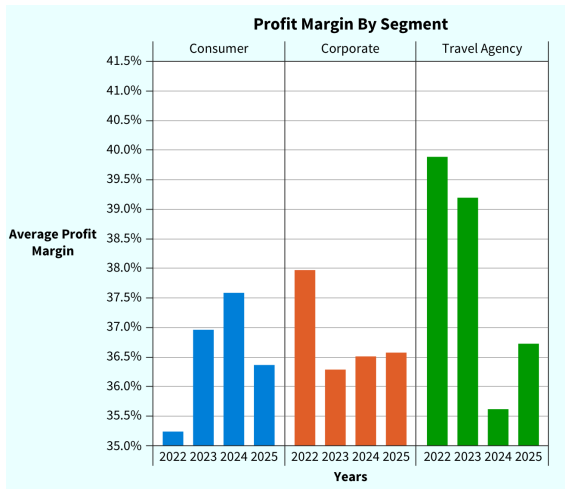
Apply It 9.3B



Madison public library 9.3

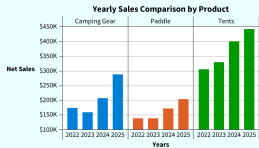
Metric:	2025	2024	2023	2022
Checkouts	3,454,156	3,575,215	3,698,903	3,800,000
Digital Checkouts	462,416	382,068		289,309
Internet Uses	227,370	247,129	564,787	635,363
Library Card Registrations	15,544	12,154	11,775	13,245
Library Employees	137	135	131	128
Meeting Room Uses	22,714	22,278	23,010	20,782
Program Attendance (all ages)	107,447	136,303	134,666	110,744
Visits	1,779,552	1,911,287	1,965,014	2,170,000

Apply It 9.4A



Apply It 9.4B

2.

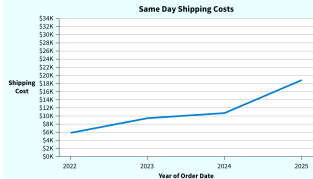


3.

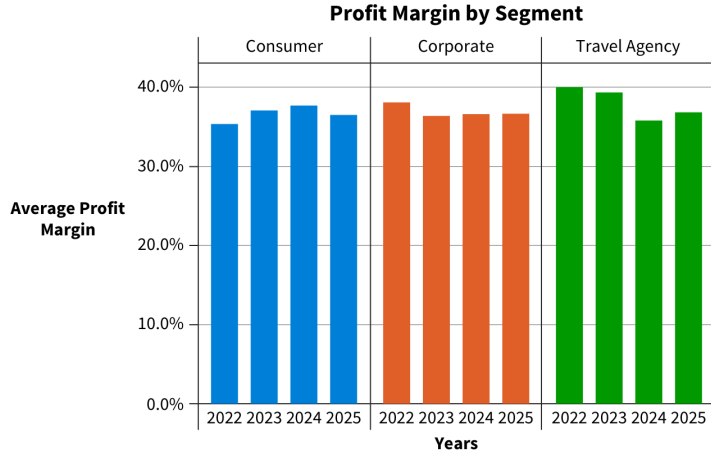
Shipping Costs by Order Priority and Shipping Mode				
Order Priority	First Class	Same Day	Second Class	Standard Class
Critical	\$ 5,905	\$ 2,664	\$ 1,654	
High	\$ 16,132	\$ 13,516	\$ 7,687	\$ 13,881
Medium	\$ 13,215	\$ 2,611	\$ 8,898	\$ 41,628
Low				\$ 5,220



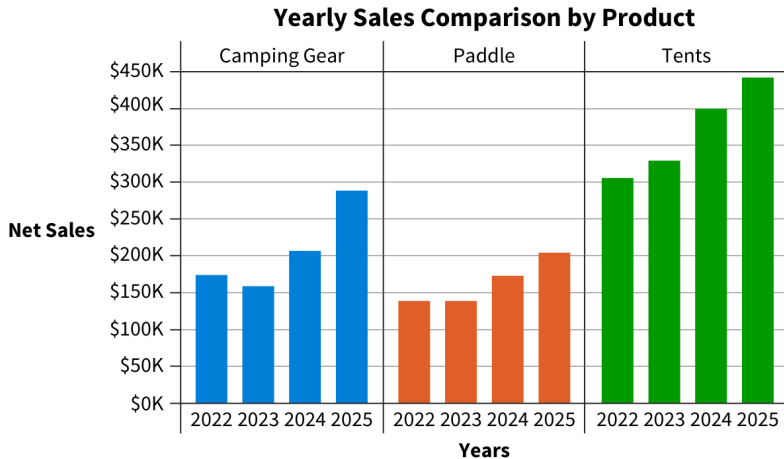
4.



Apply It 9.4C



Apply It 9.4D



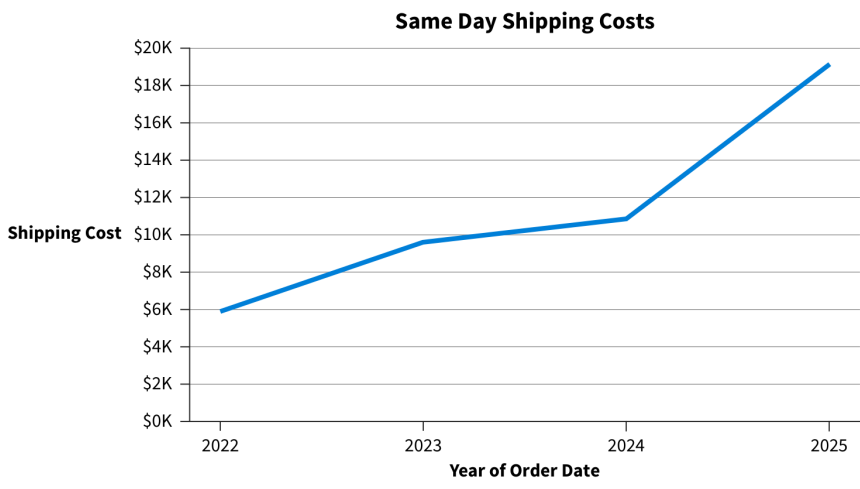
Apply It 9.4E

Shipping Cost by Mode and Order Priority: 2024–2025				
Order Priority	First Class	Same Day	Second Class	Standard Class
Critical	\$ 5,505	\$ 2,664	\$ 1,654	
High	\$ 16,132	\$ 13,516	\$ 7,687	\$ 13,881
Medium	\$ 13,215	\$ 2,611	\$ 8,898	\$ 41,628
Low				\$ 5,220

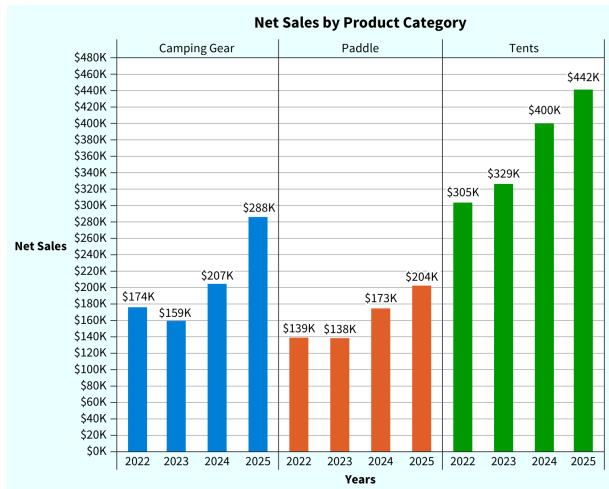
Shipping Cost



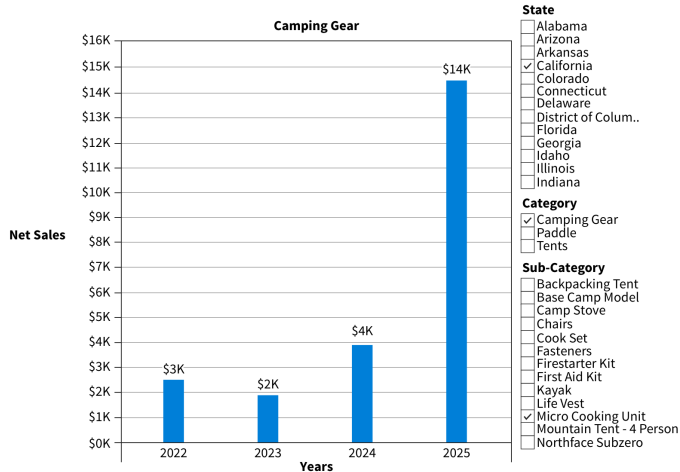
Apply It 9.4F



Apply It 9.5A



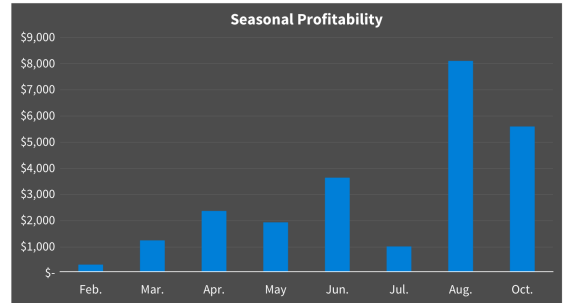
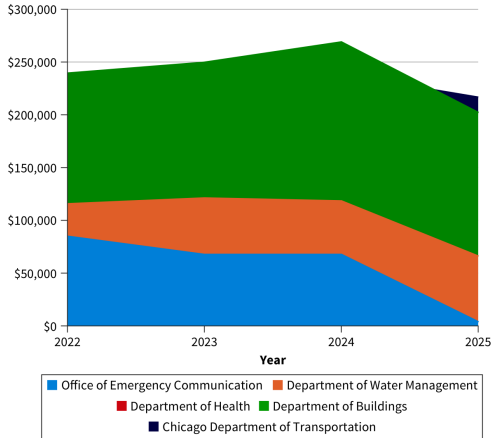
Apply It 9.5B



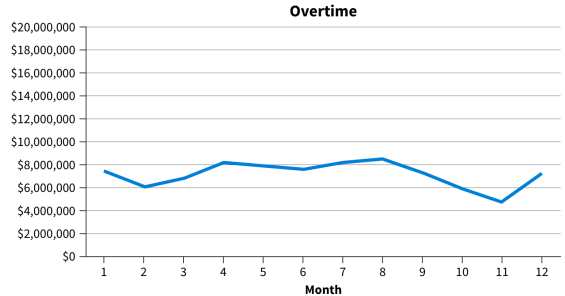
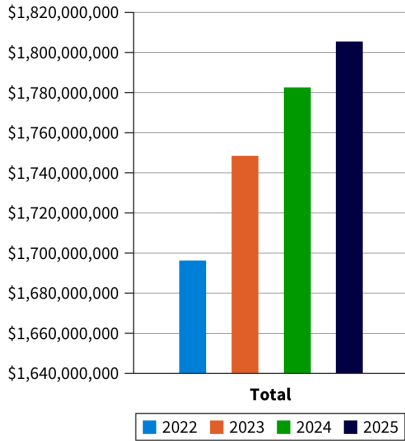
Phần Bài tập Ngắn

Brief Exercises (BE)

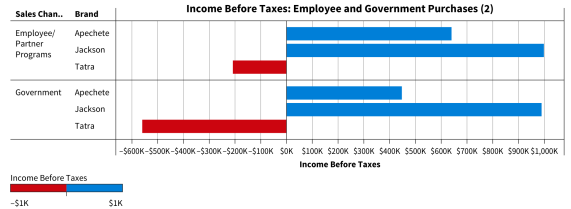
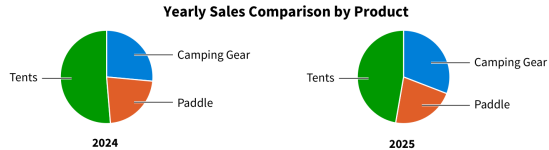
BE 9.8 & BE 9.9



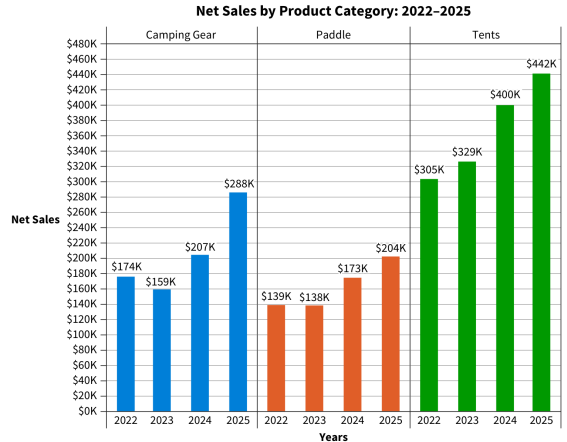
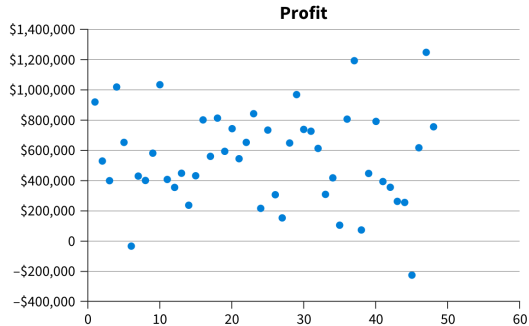
BE 9.10 & BE 9.11



BE 9.12 & BE 9.13



BE 9.14 & BE 9.15



Phần Bài tập

Exercises (EX)

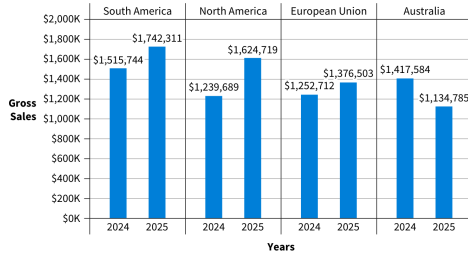
EX 9.10 & EX 9.11

Sales by Location

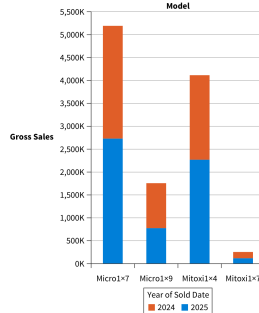
Location	Sold Date	
	2024	2025
Australia	\$ 1,417,584	\$ 1,134,785
European Union	\$ 1,252,712	\$ 1,376,503
North America	\$ 1,239,689	\$ 1,624,719
South America	\$ 1,515,744	\$ 1,742,311
Grand Total	\$ 5,425,729	\$ 5,878,318

Sales by Location

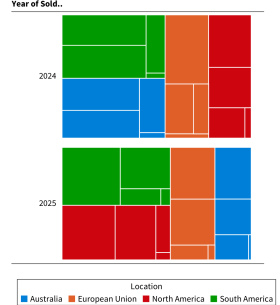
Location/Sold Date



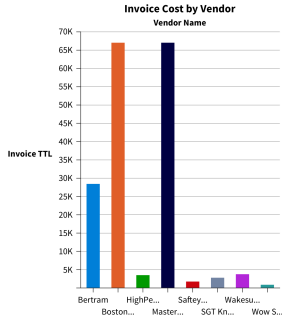
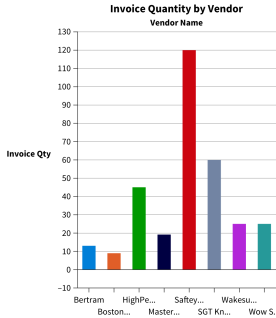
Gross Sales by Year



Sales by Model, Region, and Year

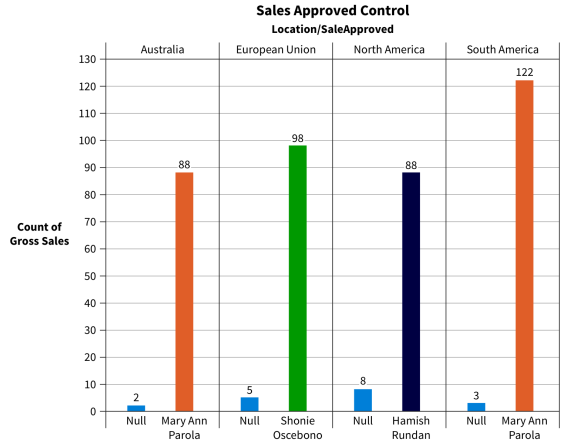


EX 9.16 & EX 9.17

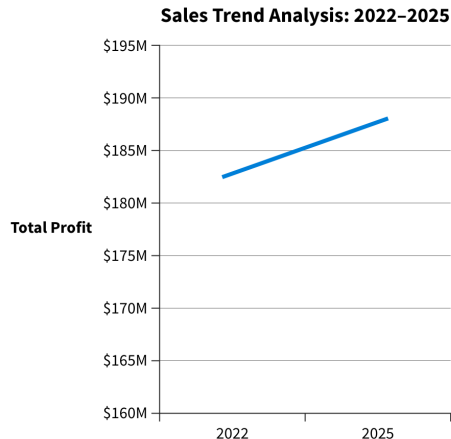
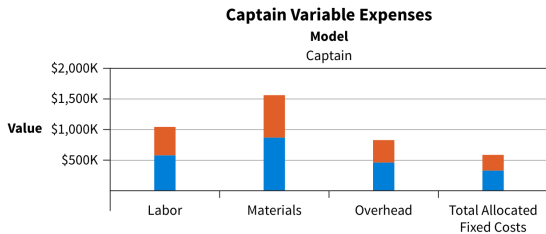


Vendor Name

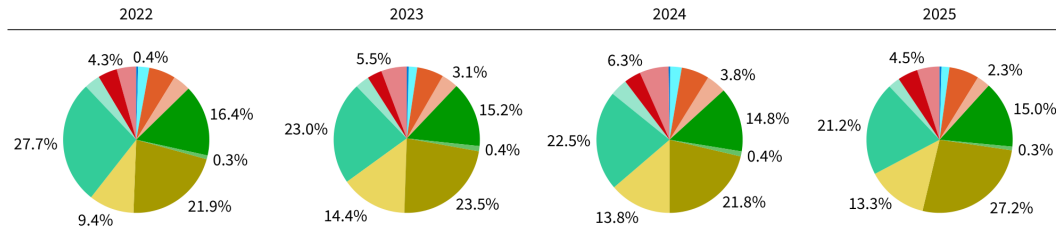
■ Bertram ■ Boston Whaler ■ HighPerforman... ■ MasterCraft ■ Saftey First ■ SGT Knots ■ Wakesurf, Inc. ■ Wow Sports, Inc.



EX 9.18 & EX 9.20A



Product Sales Mix: 2022–2025



Product Type Description



Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!